

# TFPT — Safeguards against Coincidence and Numerology

The verification discipline: how every load-bearing claim is defended against chance, fitting and over-reading

Stefan Hamann

Alessandro Rizzo

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## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions** (plus two standalone working notes), best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R. tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ) as numbered contracts; **H. tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S. tfpt\_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the fire-wall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P. tfpt\_prime\_front** — research documentation of the prime / zeta line: the seven load-bearing modules on the  $E_8$  glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

**Working notes:** **N1. note\_e8\_gaussian\_code** — the Gaussian code bridge:  $E_8$  over  $\mathbb{Z}[i]$  via Construction A over the extended Hamming code  $[8, 4, 4]$ , the canonical four-bit quotient  $L/(1+i)L \cong \mathbb{F}_2^4$ , and the  $G_{31}$  quartic companion (exact algebra, machine-certified); **N2. note\_hilbert\_polya\_truncations** — a computable, zeta-free truncation family for the Weil measure: measurements on a Hilbert–Pólya candidate (prediction-freeze methodology; no RH claim).

*You are reading the **Safeguards** companion (S): it collects, in one place and in one language, every mechanism by which TFPT protects a claim against coincidence — the status calculus, the anti-fitting rules, the over-determination accounting, the firewall, the frozen predictions and null model, the two independent proof paths (Wolfram, Lean), the red team, and the honest residual. It introduces no new physics; it makes the discipline auditable.*

## Abstract

A two-input theory that reads out many small integers is, a priori, at risk of being elaborate numerology. TFPT answers that risk not with rhetoric but with a *layered, machine-checked*

*discipline*, and this companion states all of it uniformly. The layers are: (1) a four-class status calculus with a single-source ledger and a sync audit that forbids a conditional claim from being rendered as exact; (2) an anti-fitting rule (“no free pattern”) plus a reverse audit that counts how much structure carries *no* readout; (3) an over-determination map that honestly separates *multiplicative* evidence from *compression* (many readouts from one generator) and, applied to TFPT’s own arithmetic witnesses, finds them to be facets of *one*  $(2, 3, 5)/E_8$  object (compression), locating the genuine multiplication in an input forced four independent ways plus a foreign readout ( $\alpha^{-1}$ ); (4) a firewall that keeps external-physics interfaces from ever being typed as compiler outputs, together with the No-Unit theorem that makes the absence of an absolute scale a *theorem*; (5) a frozen prediction registry with a Monte-Carlo null model and a live data scorecard; (6) two independent reproduction paths — an independent Wolfram engine and a Lean 4 kernel proof; (7) an adversarial red-team layer; and a final, explicit statement of what would still falsify the theory. The thesis is deliberately modest: these safeguards make coincidence an *expensive* explanation of the discrete core, and they keep “exact compiler closure” from ever being mistaken for “closed physics”.

## 1 The risk, and the shape of the answer

The honest danger for TFPT is not that an individual number is wrong — the arithmetic is usually clean — but *closure inflation*: that an exact, discrete *compiler* closure is read, or worded, as a *physical* closure. Every safeguard below exists to hold that one line. None of them, alone, turns syntax into semantics; together they make the discrete core hard to dismiss as chance and impossible to oversell as a finished Theory of Everything. We state each as a *layer* with its enforcing artefact(s).

The layered defense — a load-bearing claim must pass all seven

|                                                                                              |
|----------------------------------------------------------------------------------------------|
| <b>1 Status calculus</b><br>[E]/[C]/[O]/[X] · ledger single source · audit_sync              |
| <b>2 No free pattern + reverse audit</b><br>v305 · E8.REVERSE.AUDIT (3/8 readout)            |
| <b>3 Over-determination map</b><br>v427/v428 · seven readouts compress one (2,3,5)/E8 object |
| <b>4 Firewall + No-Unit</b><br>v187/v213 · v153 (v_geo theorem-forbidden)                    |
| <b>5 Frozen registry + null model</b><br>v84 · v100 Monte-Carlo · v375 scorecard             |
| <b>6 Two independent paths</b><br>Wolfram 116+327 · Lean 4 kernel proof                      |
| <b>7 Red team</b><br>tfpt_5 · Targets A-F · named kill tests                                 |

Figure 1: The seven safeguard layers; a load-bearing claim must pass all of them. Each layer names its enforcing artefact (script id or document).

## 2 Layer 1 — the status calculus (no [C] dressed as [E])

Every claim carries one of four display markers: **[E]** exact / proven (identity, Lie/lattice theorem, Lean-formalised, or numerical fixed point), **[C]** conditional (physical, under named hypotheses), **[O]** open / axiom / anchor, **[X]** falsifiable kill test. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) lives in the *single source of truth*, `verification/status_ledger.csv`; the papers and the website only *mirror* it. A sync audit

(`verification/audit_sync.py`, run as `bash build.sh audit`) enforces, in both directions, that the suite, the runner, the registry and the ledger agree; that every script is cited in a paper *body*; and that no generated surface is stale. The point of the calculus is exactly the reviewer’s concern: it is structurally impossible for a [C] claim to be silently rendered as [E], because the marker is read from the ledger, not retyped per document.

### 3 Layer 2 — no free pattern, and the reverse audit

#### Anti-fitting: an identity earns its keep or it is decorative

The *forward discipline* / generator-economy audit ([`verification/v305_witness_independence.py`], null model [`verification/v100_numerology_null_mc.py`]) admits an identity as load-bearing only if it is derivationally necessary, kills alternatives, is ablation-relevant, links modules, or is experimentally testable; admissible invariant classes are bounded in advance. The complementary *reverse audit* (E8.REVERSE.AUDIT.01) asks the honest opposite question — how much  $E_8$  structure maps onto *nothing*? Of the eight Casimir degrees {2, 8, 12, 14, 18, 20, 24, 30}, exactly **three** feed a primary physical readout (degree 2, the metric; degree 8, the rank  $\rightarrow c_3 = 1/(8\pi)$ ; degree 30, the Coxeter number  $\rightarrow g_{\text{car}} = 5$  and the order-30 clock) and **five** carry no primary readout. A theory that publishes its 5/8 unused overhead is not cherry-picking — and those five are not diffuse slack but the forced two-family ladder  $6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup \text{det-ladder}\{8, 14, 20\}$  (v431), so even the overhead is structured, not mined.

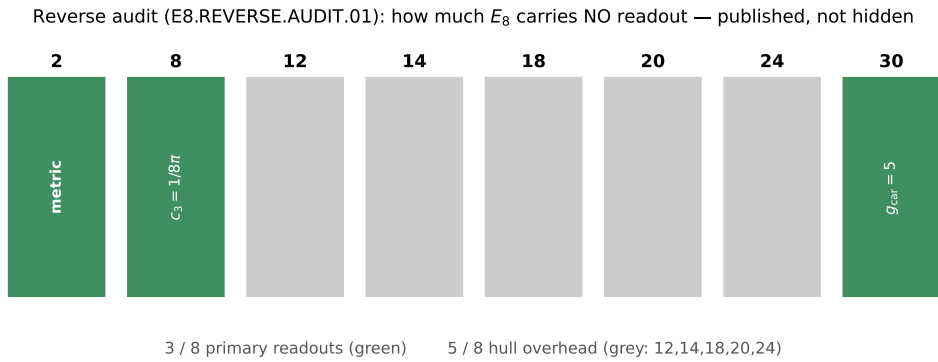


Figure 2: The reverse audit (E8.REVERSE.AUDIT.01): of  $E_8$ ’s eight Casimir degrees only three feed a primary readout; the other five are published hull overhead.

### 4 Layer 3 — the over-determination map (multiply vs. compress)

The strongest positive evidence is over-determination, but it must be counted honestly. The *framework* is a sharp axis ([`verification/v427_overdet_witness_map.py`], extending the generator-economy audit v305): evidence *multiplies* only across genuinely disjoint grammars; many readouts from one generator are a *compression* gain, not independent multiplication. The hard part is classifying TFPT’s own witnesses correctly — and applying that test to ourselves ([`verification/v428_overdet_reclass.py`]) forces a self-correction that we state in full.

### Seven arithmetic readouts — and why they compress

Each integer below is computed *from its own grammar* and lands on a carrier-skeleton value:

|                                   |                                                              |                          |
|-----------------------------------|--------------------------------------------------------------|--------------------------|
| Gauss $\mathbb{Z}[i]$ :           | $N(3+2i) = 3^2+2^2$                                          | $= 13 (= \Delta_Q)$      |
| Eisenstein $\mathbb{Z}[\omega]$ : | $N(3+2\omega) = 3^2-3\cdot 2+2^2$                            | $= 7$                    |
| Cyclotomy $\mathbb{Q}(\zeta_5)$ : | $N(3+2\zeta_5) = 2^4 \Phi_5(-3/2)$                           | $= 55$                   |
| Galois $(\mathbb{Z}/5)^\times$ :  | $ \text{Gal } \mathbb{Q}(\zeta_5)  = \varphi(5)$             | $= 4 (=  \mu_4 )$        |
| Root lattice :                    | $ \det \text{Cartan}(E_8) $                                  | $= 1$                    |
| Pascal/exterior :                 | $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 11, \quad 2^4$ | $= 16 = \dim S^+$        |
| Coxeter :                         | $h(E_8) = 30 = 2\cdot 3\cdot 5, \quad \varphi(30)$           | $= 8 = \text{rank } E_8$ |

These are seven *syntactically* distinct grammars — but syntactic distinctness is not statistical independence. By the Brieskorn classification (v236) the  $(2, 3, 5)$  singularity  $x^2+y^3+z^5$  is the *one* generator behind the order-30 clock, with Milnor number  $\mu = (2-1)(3-1)(5-1) = 8 = \text{rank } E_8$  and  $30 = 2\cdot 3\cdot 5 = h(E_8)$ . Each readout above is a facet of that same object:  $\mathbb{Z}[i]$  is the prime-2 ( $\mu_4$ /sheet) face,  $\mathbb{Z}[\omega]$  the prime-3 (family) face,  $\mathbb{Q}(\zeta_5)$  and the Galois group the prime-5 (carrier) face, Pascal the  $\mu_4$  exterior, the Coxeter number the whole clock, and  $\det \text{Cartan}(E_8)$  the  $E_8$  resolution itself. *So the seven do not multiply — they compress*, exactly like the anchor  $a = (1, 1, 2)$  emitting  $(e_1, e_2, e_3) = (4, 5, 2)$ , the power sums, and the  $E_8$  data from one generator. An earlier version of this section called the seven “disjoint grammars that multiply”; [verification/v428\_overdet\_reclass.py] applies this layer’s own test to that claim and walks it back.

### What honestly multiplies: forced inputs + a foreign witness

The genuine over-determination is of a different shape. First, the *input* of the theory is forced by several *independent* arguments: the “8” in  $c_3 = 1/(8\pi)$  equals  $\text{rank } E_8 = 8$ , the carrier Coxeter number  $h(D_5) = 2(5-1) = 8$ , the totient  $\varphi(30) = 8$ , and the Milnor number  $(2-1)(3-1)(5-1) = 8$  — four *different* mathematical facts that independently fix the same constant, and that genuinely multiply. Second, the constant the theory was *not* built around —  $\alpha^{-1} \approx 137$ , a prime foreign to the  $(2, 3, 5)$  skeleton — is reproduced as the fixed point of the boundary Ward identity (v305). The honest claim is therefore narrower and harder to attack: not “seven independent witnesses,” but *one richly over-determined input plus a foreign readout*.

### The unconditional floor, and the $\alpha$ -form lever [C] ([verification/v432\_overdet\_floor.py])

Multiplying *only* across the genuinely disjoint pieces — the input over-determination (the “8” forced four ways), the foreign  $\alpha^{-1}$  census ( $\sim 1/94500$ , v100), the  $\varphi_0$ -seed cluster counted *once*, and one downstream witness ( $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ ) — yields an *unconditional* improbability floor:  $\leq 10^{-6}$  across a grid of deliberately generous chance assignments, nominal  $\sim 10^{-10}$ . This floor is grammar-*independent* (it survives even if a skeptic rejects the declared formula grammar) and sits  $\sim 20$  orders *above* the conditional null-model ceiling  $10^{-30.7}$  (v100). That gap is exactly the evidential payoff of *deriving* the  $\alpha$  form: the open form-derivation ALPHA.QUILLEN.EXACT.01 (v382; its cubic level now the *computed* bulk Chern invariant  $|C|=1$  and its normalisation the embedding index  $k_Y=5/3$ , v470; the determinant-line bridge lemma exhibited at the finite level over the  $U(1)$ -twist moduli, v472 — the continuum  $\zeta$ -det identification still [O]) and this floor are the *same lever* — closing it moves the  $\alpha$  factor from the  $1/94500$  census toward certainty. Honest scope: this is a conservative *floor* (an upper bound on chance-plausibility), not a posterior, and it bounds only the “is it numerology?” question — not the physics bridge (the firewall, v187). *Hardened* ([verification/v436\_overdet\_floor2.py]). The verdict does not even need the chance assignments above: stripped to a *pure counting* census — of the 94500 complexity-matched  $F_{U(1)}$  variants only the TFPT one hits the CODATA window (v100) — the floor is  $1/94500 \approx 4.40\sigma$  with *no* subjective probability, prior or multi-observable grammar. A monotone concession ladder ( $30.7 \rightarrow 25.8 \rightarrow 4.98$  dex) keeps the “not chance” verdict at the *most adversarial* rung, so it is independent of the contestable input-“8” independence; the only honest gap to the conventional  $5\sigma$  line is exactly one further independent confirmation (a factor  $\leq 0.054$ ), stated openly.

## 5 Layer 4 — the firewall and the No-Unit theorem

Two safeguards keep the boundary between the closed core and external physics sharp.

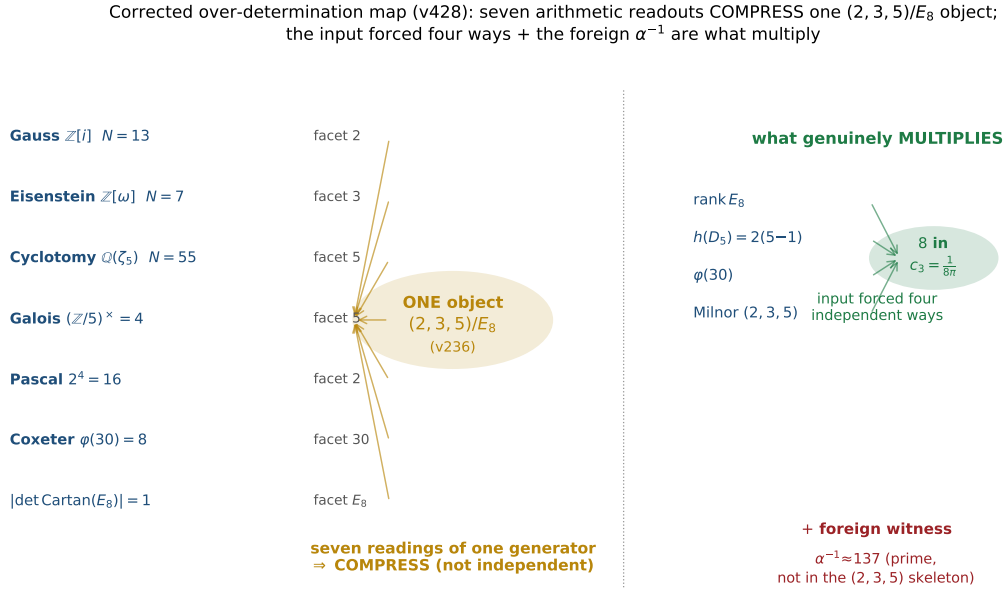


Figure 3: The corrected over-determination map ([verification/v428\_overdet\_reclass.py]): the seven arithmetic readouts are facets of *one*  $(2, 3, 5)/E_8$  object (a *compression* gain, like the anchor), so they do not multiply; what genuinely multiplies is the input forced four independent ways (the “8”) plus the foreign witness  $\alpha^{-1} \approx 137$ . The disjoint pieces give an *unconditional* floor  $\sim 10^{-10}$  ([verification/v432\_overdet\_floor.py]),  $\sim 20$  orders above the conditional  $10^{-30.7}$ ; hardened to an assumption-minimal *counting* floor  $1/94\,500 \approx 4.40\sigma$  ([verification/v436\_overdet\_floor2.py]) that drops every subjective chance assignment and a monotone concession ladder ( $30.7 \rightarrow 25.8 \rightarrow 4.98$  dex) that keeps the “not chance” verdict at the most adversarial rung.

- **The firewall.** The four frontier transfers —  $F_{\text{pole}}$  (Koide),  $F_{\text{Boltzmann}}$  ( $\eta_B$ ),  $F_{\text{relic}}$  (axion),  $F_{\text{QCD}}$  ( $m_p/m_e$ ) — are *typed interfaces*, a discrete compiler kernel composed with an external continuous solver. A guard ([verification/v187\_ftransfer\_laws.py]) reads the ledger and *enforces* that none is ever typed as a primitive [E] compiler prediction; the functor contract is [verification/v213\_ftransfer\_functor.py] and a prose sentinel ([verification/v188\_frontier\_wording\_guard.py]) guards the wording. Exact algebraic sub-parts may be [E] (the Koide 53/54 factor,  $b_3 = -7$ ); the physical observable never is. The recent single-flow reduction ([verification/v425\_dyn\_transfer\_universal.py]) makes the *dynamics* of all four the one native seam recovery semigroup, leaving only named anchors external — a sharpening that respects, not relaxes, the firewall.
- **The No-Unit theorem** ([verification/v153\_no\_unit\_theorem.py]). Because  $c_3$  and  $g_{\text{car}}$  are mass-dimension 0, no dimensionless compiler can output an absolute scale;  $v_{\text{geo}}$  is therefore *forbidden by theorem*, not a missing prediction. The residual-certification audit ([verification/v384\_residual\_certification.py]) makes the shape explicit: every open item is an external math proof, theorem-forbidden (the unit), or external physics — *zero* open internal mechanisms.

## 6 Layer 5 — frozen predictions, the null model, and the $\alpha$ uniqueness scan

Predictions are frozen before confrontation, and a match is scored against *chance*, not admired in isolation.

**The look-elsewhere null model, made explicit ([verification/v100\_numerology\_null\_mc.py])**

The frozen registry ([verification/v84\_frozen\_registry.py], REG.FREEZE.01) fixes every dimensionless prediction at 25 digits, re-derived from the two axioms each run (a formula $\leftrightarrow$ value lock). The null model then runs an *exact census* of the full declared formula grammar over the 13 scored observables with conservative data windows: the joint formula-fishing probability is  $\prod_i p_i = 10^{-25.8}$ , and  $2 \times 200,000$  Monte-Carlo pseudo-theories reach *at most* 5/13 hits against TFPT's 13/13 (Fig. 4). Power controls confirm the test bites:  $\varphi_0^{\text{ret}} \pm 10\%$  collapses TFPT's own scorecard to 6/13, and a data shuffle averages 1.2/13.

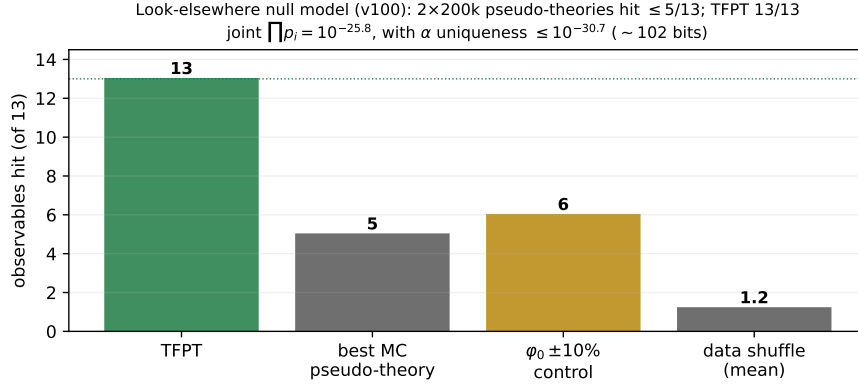


Figure 4: The look-elsewhere null model ([verification/v100\_numerology\_null\_mc.py]): Monte-Carlo pseudo-theories hit at most 5/13 where TFPT hits 13/13; the power controls ( $\varphi_0^{\text{ret}} \pm 10\%$ , data shuffle) confirm the test is non-vacuous.

**Is the  $\alpha$  path unique?** The sharpest single result is the fine-structure fixed point. All 94,500 variants of the boundary  $U(1)$  Ward formula  $F_{U(1)}$  in the declared grammar are root-solved; *exactly one* — the budget  $M=41$  at  $N=8$  ( $c_3 = 1/8\pi$ ) — lands in the  $4 \times 10^{-8}$  CODATA window (Fig. 5). Folding this into the census gives a total  $\leq 10^{-30.7}$  ( $\sim 102$  bits), *conditional on the declared grammar* — a null-model rejection, never a claim of certainty.

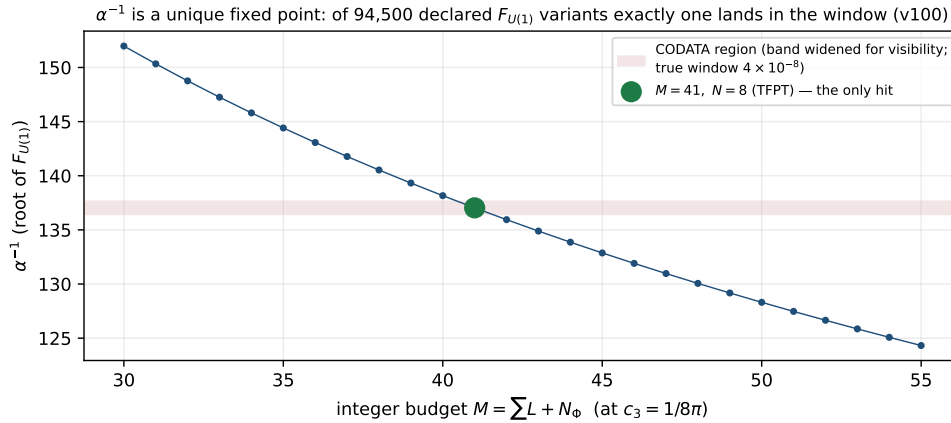


Figure 5:  $\alpha^{-1}$  as a unique fixed point: scanning the integer budget  $M$  at  $c_3 = 1/8\pi$ , only  $M=41$  lands in the CODATA window; the full census over all 94,500  $F_{U(1)}$  variants finds exactly one hit.

**No moving goalposts; the worst case is shown.** A live data scorecard records the JUNO/NuFIT/ACT/BK18 compatibility ([verification/v375\_observatory\_registry.py]). A data watchdog types each row as PASS, WATCH, TENSION or KILL ([verification/v307\_data\_watchdog.py]), and a kill-test board keeps the falsifiers tied to the frozen rows ([verification/v321\_killtest\_board.py]).

The discipline is applied to its own weakest point: the reactor angle  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$  is *pre-registered* at  $+2.0\sigma$  tension (FLAV.TH13.PRESSURE.01) with its own kill criterion — the opposite of hiding the worst case.

## 7 Layer 6 — two independent reproduction paths

A single implementation can share a single bug; TFPT therefore re-derives the exact core twice more, independently.

- **An independent Wolfram engine** (GATE.WOLFRAM.01/GATE.WOLFRAM.02): 116/116 base checks plus a 588/588 extension mirror every exact algebraic / identity / lattice result in a different computer-algebra system, ending ALL WOLFRAM CHECKS PASSED. (Numerical ODE/PDE results are Python-only by design and flagged as such.)
- **A Lean 4 kernel proof**: the load-bearing algebra is machine-formalised with no *sorry* — the carrier hypercharge as the exact anomaly-free root of  $6Y^2 - Y - 1 = 0$ , the Pascal carrier ladder  $2^4 = \binom{5}{0} + \binom{5}{1} + \binom{5}{2}$  (FORM.LADDER.01), the integer rigidity of the carrier, the seam-equivalence chain (FORM.SEAMEQUIV.01) and its collar/MMST face (FORM.SEAM.MMST.01) — so the kernel, not the prose, certifies the core.

## 8 Layer 7 — the red team

The adversarial audit (companion `tfpt_5_redteam`) does not defend; it *attacks*. It declares six targets — A (the seam =  $(E_8)_1$  identification), B (the carrier truncation  $g_{\text{car}}=5$ ), C (the  $k=c_3/2$  area law), D (the ratios  $\rightarrow v_{\text{geo}}$  reduction), E (whether one scale carries the theory), F (the perturbative 4D-QFT round) — states what survives each, and reduces every survivor to a named residual. The model example is its own honesty about holomorphy:  $c = 8$  alone does *not* select  $(E_8)_1$  because  $(D_8)_1 = SO(16)_1$  shares  $c = 8$ ; the load-bearing extra is holomorphy ( $|\det K| = 1$ ). Publishing the sharpest attack on oneself is the strongest anti-numerology signal a programme can give.

## 9 What would still falsify TFPT

The safeguards are only credible with explicit kill tests (companion “How to kill TFPT”). Among the frozen ones: a fourth chiral generation (breaks  $N_{\text{fam}} = 3$ ); a genuinely free fundamental constant that is not an  $E_8$ -closure fixed point; a seam denominator  $\neq 8$ ;  $\alpha^{-1}$  away from the fixed-point root; a stable  $\sin^2 \theta_{13}$  far from 0.0231 at  $> 3\sigma$ ; cosmic birefringence converging away from  $\beta \approx 0.2424^\circ$ ; a tensor-to-scalar  $r \gtrsim 0.01$  (which kills the  $R^2$  branch). The honest residual above all kill tests is exactly three named interfaces — the seam continuum theorem (SEAM.EQUIV.01, now reduced to a cited commuting-projector result, [\[verification/v424\\_seam\\_lto\\_rp\\_reduction.py\]](#)/[\[verification/v426\\_seam\\_lto\\_rp\\_kms.py\]](#), its extension leg since re-founded on peer-reviewed crossed-product theorems with the realisation input at invariant level, [\[verification/v469\\_seam\\_crossedproduct\\_route.py\]](#)), the metrology unit  $v_{\text{geo}}$  (theorem-forbidden), and the typed  $F_{\text{transfer}}$  interfaces (firewalled).

## 10 Summary — the layered defense

| Layer                           | What it defends against                                            | Artefact                                        |
|---------------------------------|--------------------------------------------------------------------|-------------------------------------------------|
| Status calculus                 | a <b>[C]</b> claim read as <b>[E]</b>                              | ledger + <code>audit_sync.py</code>             |
| No free pattern + reverse audit | cherry-picking, decorative identities                              | <code>v305/v100/v431/E8.REVERSE.AUDIT.01</code> |
| Over-determination map          | rhetorical “everything multiplies” (incl. <i>our own</i> miscount) | <code>v427/v428/v432/v436</code>                |
| Firewall + No-Unit              | external physics sold as compiler output; a faked scale            | <code>v187/v213/v153/v384</code>                |
| Frozen registry + null model    | moving goalposts; chance hits                                      | <code>v84/v100/v375</code>                      |
| Two independent paths           | a shared implementation bug                                        | Wolfram + Lean                                  |
| Red team                        | self-serving framing                                               | <code>tfpt_5_redteam</code>                     |

The claim defended here is deliberately narrow: *coincidence is an expensive explanation of the discrete core, and exact compiler closure is never silently upgraded to physical closure*. That is the whole job of this companion.